

Introduction to Operations Management

- **HAZEL**

Hazel had worked for the same *Fortune* 500 company for almost 15 years. Although the company had gone through some tough times, things were starting to turn around. Customer orders were up, and quality and productivity had improved dramatically from what they had been only a few years earlier due to a companywide quality improvement program. Thus, it came as a real shock to Hazel and about 400 of her coworkers when they were suddenly terminated following the new CEO's decision to downsize the company.

After recovering from the initial shock, Hazel tried to find employment elsewhere. Despite her efforts, after eight months of searching she was no closer to finding a job than the day she started. Her funds were being depleted and she was getting more discouraged. There was one bright spot, though: She was able to bring in a little money by mowing lawns for her neighbors. She got involved quite by chance when she heard one neighbor remark that now that his children were on their own, nobody was around to cut the grass. Almost jokingly, Hazel asked him how much he'd be willing to pay. Soon Hazel was mowing the lawns of five neighbors. Other neighbors wanted her to work on their lawns, but she didn't feel that she could spare any more time from her job search.

However, as the rejection letters began to pile up, Hazel knew she had to decide. On a sunny Tuesday morning, she decided, like many others in a similar situation, to go into business for herself—taking care of neighborhood lawns. She was relieved to give up the stress of job hunting, and she was excited about the prospect of being her own boss. But she was also fearful of being completely on her own. Nevertheless, Hazel was determined to make a go of it.

At first, business was a little slow, but once people realized Hazel was available, many asked her to take care of their lawns. Some people were simply glad to turn the work over to her; others switched from professional lawn care services. By the end of her first year in business, Hazel knew she could earn a living this way. She also performed other services such as fertilizing lawns, weeding gardens, and trimming shrubbery. Business became so good that Hazel hired two part-time workers to assist her and, even then, she believed she could expand further if she wanted to.

• **Question**

1. Hazel is the operations manager of her business. Among her responsibilities are forecasting, inventory management, scheduling, quality assurance, and maintenance.^{[1][SEP]}
 - a. What kinds of things would likely require forecasts?
 - b. What inventory items does Hazel probably have? Name one inventory decision she must make periodically.
 - c. What scheduling must she do? What things might occur to disrupt schedules and cause Hazel to reschedule?
 - d. How important is quality assurance to Hazel's business? Explain.
 - e. What kinds of maintenance must be performed?^{[1][SEP]}
2. In what ways are Hazel's customers most likely to judge the quality of her lawn care services?
3. What are some of the trade-offs that Hazel probably considered relative to:^{[1][SEP]}
 - a. Working for a company instead of for herself?
 - b. Expanding the business?^{[1][SEP]}
 - c. Launching a website?
4. The town is considering an ordinance that would prohibit putting grass clippings at the curb for pickup because local landfills cannot handle the volume. What options might Hazel consider if the ordinance is passed? Name two advantages and two drawbacks of each option.
5. Hazel decided to offer the students who worked for her a bonus of \$25 for ideas on how to improve the business, and they provided several good ideas. One idea that she initially rejected now appears to hold great promise. The student who proposed the idea has left and is currently working for a competitor. Should Hazel send that student a check for the idea? What are the possible trade-offs?
6. All managers have to cope with variation.^{[1][SEP]}
 - a. What are the major sources of variation that Hazel has to contend with?^{[1][SEP]}
 - b. How might these sources of variation impact Hazel's ability to match supply and demand?^{[1][SEP]}
 - c. What are some ways she can cope with variation?
7. Hazel is thinking of making some of her operations sustainable. What are some ideas she might consider?
8. What competitive advantage does Hazel have over a professional lawn care service?^{[1][SEP]}
9. Hazel would like to increase her profits, but she doesn't believe that it would be wise to raise her prices considering the current state of the local economy. Instead, she has given some thought to increasing productivity.^{[1][SEP]}
 - a. Explain how increased productivity an alternative could be to increased prices.^{[1][SEP]}
 - b. What are some ways that Hazel could increase productivity?^{[1][SEP]}
10. Hazel is thinking about the purchase of new equipment. One would be power sidewalk edgers. She believes edgers will lead to an increase in productivity. Another would be a chain^{[1][SEP]}

11. Hazel has been successful in her neighborhood, and now wants to expand to other neighborhoods, including some that are five miles away. What would be the advantages and disadvantages of doing this?

12. Hazel does not have a mission statement or a set of objectives. Take one of the following positions and defend it:

- a. Hazel doesn't need a formal mission statement and ^{{}{}}_{SEP} objectives. Many small businesses don't have them. ^{{}{}}_{SEP}
- b. She needs a mission statement and a set of objectives. They would be extremely beneficial. ^{{}{}}_{SEP}
- c. There may be some benefit to Hazel's business, and she ^{{}{}}_{SEP} should consider developing one. ^{{}{}}_{SEP}

Competitiveness, Strategy, and Productivity

• AN AMERICAN TRAGEDY: How A GOOD COMPANY DIED

ZACHARY SCHILLER

The Rust Belt is back. So say bullish observers as U.S. exports surge, long-moribund industries glow with newfound profits, and unemployment dips to lows not seen in a decade. But in the smokestack citadels, there's disquiet. Too many machine-tool and auto parts factories are silent; too many U.S. industries still can't hold their own. What went wrong since the heyday of the 1960s? That's the issue Max Holland, a contributing editor of *The Nation*, takes up in his nutsy-boltsy but fascinating study, *When the Machine Stopped*.*

The focus of the story is Burgmaster Corp., a Los Angeles-area machine-tool maker founded in 1944 by Czechoslovakian immigrant Fred Burg. Holland's father worked there for 29 years, and the author interviewed 22 former employees. His shop-floor view of this small company is a refreshing change from academic treatises on why America can't compete.

The discussions of spindles and numerical control can be tough going. But Holland compensates by conveying the excitement and innovation of the company's early days and the disgust and cynicism accompanying its decline. Moreover, the fate of Burgmaster and its brethren is crucial to the U.S. industrial economy: Any manufactured item is either made by a machine tool or by a machine made by a machine tool.

Producing innovative turret drills used in a wide variety of metal working tasks, Burgmaster was a thriving enterprise by 1965, when annual sales amounted to about \$8 million. The company needed backing to expand, however, so it sold out to Buffalo-based conglomerate Houdaille Industries Inc. Houdaille was in turn purchased in a 1979 leveraged buyout (LBO) led by Kohlberg Kravis Roberts & Co. By 1982, when debt, competition, and a sickly machine-tool market had battered Burgmaster badly, Houdaille went to Washington with a petition to withhold the investment tax credit for certain Japanese-made machine tools.

Thanks to deft lobbying, the Senate passed a resolution supporting Houdaille's position, but President Reagan refused to go along. Houdaille's subsequent attempt to link Burgmaster up with a Japanese rival also failed, and Burgmaster was closed. Holland uses Burgmaster's demise to explore some key issues of economic and trade policy. Houdaille's charge that a cartel led by the Japanese government had injured U.S. toolmakers, for example, became a rallying point for those who would blame a fearsome Japan Inc. for the problems of U.S. industry.

Holland describes the Washington wrangling over Houdaille in painful detail. But he does show that such government decisions are often made without much knowledge of what's going on in industry. He shows, too, that Japanese producers succeeded less because of government help than because they made better, cheaper machines.

For those who see LBOs as a symptom of what ails the U.S. economy, Holland offers plenty of ammunition. He argues persuasively that the LBO crippled Burgmaster by creating enormous pressure to generate cash. As Burgmaster pushed its products out as fast as possible, he writes, it routinely shipped defective machines. It promised customers features that engineers hadn't yet designed. And although KKR disputes the claim, Holland concludes that the LBO choked off Burgmaster's investment funds just when foreign competition made them most necessary. As for Houdaille, it was recapitalized and sold to Britain's Tube Investments Group.

But Burgmaster's problems had started even before the LBO. Holland's history of the company under Houdaille is a veritable catalog of modern management techniques that flopped. One of the most disastrous was a system for computerizing production scheduling that was too crude for complex machine-tool manufacturing. Holland gives a dramatic depiction of supply snafus that resulted in delays and cost increases.

As an independent company, "Burgmaster thrived because the Burgs knew their business," Holland writes. Their departure under Houdaille was followed by an "endless and ultimately futile search for a better formula." But, he concludes: "No formula was a substitute for management involvement on the shop floor."

In the end, however, Holland puts most of the blame for the industry's decline on government policy. He targets tax laws and macroeconomic policies that encourage LBOs and speculation instead of productive investment. He also criticizes Pentagon procurement policies for favoring exotic, custom machines over standard, low-cost models. This adds up to an industrial policy, Holland writes—a bad one.

The point is well taken, but Holland gives it excessive weight. Like their brethren in Detroit and Pittsburgh, domestic toolmakers in the 1970s were too complacent when imports seized the lower end of the product line. The conservatism that had for years served them in their cyclical industry left them ill-prepared for change. Even now some of the largest U.S. tool-makers are struggling to restructure. Blame the government, yes. But blame the industry, too.

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• **Question**

1. Write a brief report that outlines the reasons (both internal and external) for Burgmaster's demise, and whether operations management played a significant role in the demise. [1] [SEP]
2. Do you think that inadequate strategic planning was a factor that resulted in the company's asking for trade protection? [1] [SEP]
3. Can you think of a strategy that could have increased Burgmaster's chance of survival? Explain why you think that strategy would have been effective. [1] [SEP]

- **Home-style Cookies**

The Company

The baking company is located in a small town in New York State. The bakery is run by two brothers. The company employs fewer than 200 people, mainly blue-collar workers, and the atmosphere is informal.

The Product

The company's only product is soft cookies, of which it makes over 50 varieties. Larger companies, such as Nabisco, Sunshine, and Keebler, have traditionally produced biscuit cookies, in which most of the water has been baked out, resulting in crisp cookies. The cookies have no additives or preservatives. The high quality of the cookies has enabled the company to develop a strong market niche for its product.

The Customers

The cookies are sold in convenience stores and supermarkets throughout New York, Connecticut, and New Jersey. The company markets its cookies as "good food"—no additives or preservatives—and this appeals to a health-conscious segment of the market. Many customers are over 45 years of age, and prefer a cookie that is soft and not too sweet. Parents with young children also buy the cookies.

The Production Process

The company has two continuous band ovens that it uses to bake the cookies. The production process is called a batch processing system. It begins as soon as management gets orders from distributors. These orders are used to schedule production. At the start of each shift, a list of the cookies to be made that day is delivered to the person in charge of mixing. That person checks a master list, which indicates the ingredients needed for each type of cookie, and enters that information into the computer. The computer then determines the amount of each ingredient needed, according to the quantity of cookies ordered, and relays that information to storage silos located outside the plant where the main ingredients (flour, sugar, and cake flour) are stored. The ingredients are automatically sent to giant mixing machines where the ingredients are combined with proper amounts of eggs, water, and flavorings. After the ingredients have been mixed, the batter is poured into a cutting machine where it is cut into individual cookies. The cookies are then dropped onto a conveyor belt and transported through one of two ovens. Filled cookies, such as apple, date, and raspberry, require an additional step for filling and folding.

The non-filled cookies are cut on a diagonal rather than round. The diagonal-cut cookies require less space than straight-cut cookies, and the result is a higher level of productivity. In addition, the company recently increased the length of each oven by 25 feet, which also increased the rate of production.

As the cookies emerge from the ovens, they are fed onto spiral cooling racks 20 feet high and 3 feet wide. As the cookies come off the cooling racks, workers place the cookies into boxes manually, removing any broken or deformed cookies in the process. The boxes are then wrapped, sealed, and labeled automatically.

Inventory

Most cookies are loaded immediately onto trucks and shipped to distributors. A small percentage are stored temporarily in the company's warehouse, but they must be shipped shortly because of their limited shelf life. Other inventory includes individual cookie boxes, shipping boxes, labels, and cellophane for wrapping. Labels are reordered frequently, in small batches, because FDA label requirements are subject to change, and the company does not want to get stuck with labels it can't use. The bulk silos are refilled two or three times a week, depending on how quickly supplies are used.

Cookies are baked in a sequence that minimizes downtime for cleaning. For instance, light-colored cookies (e.g., chocolate chip) are baked before dark-colored cookies (e.g., fudge), and oatmeal cookies are baked before oatmeal raisin cookies. This permits the company to avoid having to clean the processing equipment every time a different type of cookie is produced.

Quality

The bakery prides itself on the quality of its cookies. Cookies are sampled randomly by a quality control inspector as they come off the line to assure that their taste and consistency are satisfactory, and that they have been baked to the proper degree. Also, workers on the line are responsible for removing defective cookies when they spot them. The company has also installed an X-ray machine on the line that can detect small bits of metal filings that may have gotten into cookies during the production process. The use of automatic equipment for transporting raw materials and mixing batter has made it easier to maintain a sterile process.

Scrap

The bakery is run very efficiently and has minimal amounts of scrap. For example, if a batch is mixed improperly, it is sold for dog food. Broken cookies are used in the oatmeal cookies. These practices reduce the cost of ingredients and save on waste disposal costs. The company also uses heat reclamation: The heat that escapes from the two ovens is captured and used to boil the water that supplies the heat to the building. Also, the use of automation in the mixing process has resulted in a reduction in waste compared with the manual methods used previously.

New Products

Ideas for new products come from customers, employees, and observations of competitors' products. New ideas are first examined to determine whether the cookies can be made with existing equipment. If so, a sample run is made to determine the cost and time requirements. If the results are satisfactory, marketing tests are conducted to see if there is a demand for the product.

Potential Improvements

There are several areas of potential improvement at the bakery. One possibility would be to automate packing the cookies into boxes. Although labor costs are not high, automating the process might save some money and increase efficiency. So far, the owners have resisted making this change because they feel an obligation to the community to employ the 30 women who now do the boxing manually. Another

possible improvement would be to use suppliers who are located closer to the plant. That would reduce delivery lead times and transportation costs, but the owners are not convinced that local suppliers could provide the same good quality. Other opportunities have been proposed in recent years, but the owners rejected them because they feared that the quality of the product might suffer.

• **Question**

1. Briefly describe the cookie production process.<sup>[17]
[SEP]</sup>
2. What are two ways that the company has increased productivity? Why did increasing the length of the ovens result in a faster output rate?<sup>[17]
[SEP]</sup>
3. Do you think that the company is making the right decision by not automating the packing of cookies? Explain your reasoning. What obligation does a company have to its employees in a situation such as this? What obligation does it have to the community? Is the size of the town a factor? Would it make a difference if the company was located in a large city? Is the size of the company a factor? What if it were a much larger company?
4. What factors cause the company to carry minimal amounts of certain inventories? What benefits result from this policy?
5. As a consumer, what things do you consider in judging the quality of cookies you buy in a supermarket?
6. What advantages and what limitations stem from the company's not using preservatives in cookies?
7. Briefly describe the company's strategy.

Forecasting

- **M&L Manufacturing**

M&L Manufacturing makes various components for printers and copiers. In addition to supplying these items to a major manufacturer, the company distributes these and similar items to office supply stores and computer stores as replacement parts for printers and desktop copiers. In all, the company makes about 20 different items. The two markets (the major manufacturer and the replacement market) require somewhat different handling. For example, replacement products must be packaged individually whereas products are shipped in bulk to the major manufacturer.

The company does not use forecasts for production planning. Instead, the operations manager decides which items to produce and the batch size, based on orders and the amounts in inventory. The products that have the fewest amounts in inventory get the highest priority. Demand is uneven, and the company has experienced being overstocked on some items and out of others. Being understocked has occasionally created tensions with the managers of retail outlets. Another problem is that prices of raw materials have been creeping up, although the operations manager thinks that this might be a temporary condition.

Because of competitive pressures and falling profits, the manager has decided to undertake a number of changes. One change is to introduce more formal forecasting procedures in order to improve production planning and inventory management.

With that in mind, the manager wants to begin forecasting for two products. These products are important for several reasons. First, they account for a disproportionately large share of the company's profits. Second, the manager believes that one of these products will become increasingly important to future growth plans; and third, the other product has experienced periodic out-of-stock instances.

The manager has compiled data on product demand for the two products from order records for the previous 14 weeks. These are shown in the following table.

Week	Product 1	Product 2
1	50	40
2	54	38
3	57	41
4	60	46
5	64	42
6	67	41
7	90*	41
8	76	47
9	79	42
10	82	43
11	85	42
12	87	49
13	92	43
14	96	44

- **Questions**

1. What are some of the potential benefits of a more formalized approach to forecasting?

{ L }
{ SEP }

2. Prepare a weekly forecast for the next four weeks for each product. Briefly explain why you chose the methods you used. (*Hint:* For product 2, a simple approach, possibly some sort of naive/intuitive approach, would be preferable to a technical approach in view of the manager's disdain of more technical methods.) { L }
{ SEP }

Strategic Capacity Planning for Products and Services

• OUTSOURCING OF HOSPITAL SERVICES

Due to financial pressures that many hospitals face, the Deaconess Clinic in Billings, Montana, decided to outsource a number of services, although in somewhat different ways.

First, the hospital outsourced its cafeteria food service. Although the food service employees were hired by the outside firm, they still felt a sense of ownership of their jobs, and still felt connected to the hospital because of the family atmosphere in the kitchen and the cafeteria.

When the hospital tried the same thing with housekeeping, employee turnover became a problem. An investigation revealed that because the housekeeping employees were more isolated in their work, they lost what little feeling of being connected to the hospital they had. The problem was solved by hiring the employees back but using the outsource company to manage housekeeping.

The hospital also decided to outsource its laundry service. This time the hospital approached a rival hospital about joining it in outsourcing laundry service.

• *Questions*

1. In some instances, the outsourced service occurs in a different location, while in others it takes place inside the organization doing the outsourcing, as the food service did in this case. What advantages were there in having the outsourced work performed within the hospital? Suppose a different hospital outsourced its food service but decided not to have the work performed in-house. What might its rationale be? ^[1]_[SEP]
2. In the housekeeping situation, why not just forget about outsourcing, especially since the hospital ended up rehiring its employees anyway? ^[1]_[SEP]
3. For laundry service, what might have been the rationale for asking another hospital to join it?

Management of Quality

• CHICK-N-GRAVY DINNER LINE

The operations manager of a firm that produces frozen dinners had received numerous complaints from supermarkets about the firm's Chick-n-Gravy dinners. The manager then asked her assistant, Ann, to investigate the matter and to report her recommendations.

Ann's first task was to determine what problems were generating the complaints. The majority of complaints centered on five defects: underfilled packages, a missing label, spills/mixed items, unacceptable taste, and improperly sealed packages.

Next, she took samples of dinners from the two production lines and examined each sample, making note of any defects that she found. A summary of those results is shown in the table.

The data resulted from inspecting approximately 800 frozen dinners. What should Ann recommend to the manager?

DEFECT OBSERVED

Date	Time	Line	Underfilled	Missing Label	Spill/Mixed	Unacceptable Taste	Improperly Sealed
5/12	0900	1		√√	√	√√√	
5/12	1330	2			√√		√√
5/13	1000	2				√	√√√
5/13	1345	1	√√		√√		
5/13	1530	2		√√	√√√		√
5/14	0830	1		√√√		√√√	
5/14	1100	2	√		√		√√
5/14	1400	1			√		√
5/15	1030	1		√√√		√√√√√	
5/15	1145	2			√	√√	
5/15	1500	1	√		√		
5/16	0845	2				√√	√√
5/16	1030	1		√√√	√	√√√	
5/16	1400	1					
5/16	1545	2	√	√√√√	√	√	√√

• TIP TOP MARKETS

Tip Top Markets is a regional chain of supermarkets located in the southeastern United States. Karen Martin, manager of one of the stores, was disturbed by the large number of complaints from customers at her store, particularly on Tuesdays, so she obtained complaint records from the store's customer service desk for the last several Tuesdays.

Assume you have been asked to help analyze the data and to make recommendations for improvement. Analyze the data using a check sheet, a Pareto diagram, and run charts.

Then construct a cause-and-effect diagram for the leading category on your Pareto diagram.

On July 15, changes were implemented to reduce out-of-stock complaints, improve store maintenance, and reduce checkout lines/pricing problems. Do the results of the last two weeks reflect improvement?

Based on your analysis, prepare a list of recommendations that will address customer complaints.

June 1

out of orange yogurt	produce not fresh ^[SEP]
bread stale ^[SEP]	lemon yogurt past sell date
checkout lines too long	couldn't find rice ^[SEP]
overcharged	milk past sell date ^[SEP]
double charged ^[SEP]	stock clerk rude ^[SEP]
meat smelled strange ^[SEP]	cashier not friendly ^[SEP]
charged for item not purchased	out of maple walnut ice cream
couldn't find the sponges ^[SEP]	something green in meat
meat tasted strange	didn't like music ^[SEP]
store too cold ^[SEP]	checkout lines too slow
light out in parking lot	

June 8

fish smelled funny ^[SEP]	undercharged ^[SEP]
out of diet bread	out of roses ^[SEP]
dented can ^[SEP]	meat spoiled
out of hamburger rolls	overcharged on two items
fish not fresh	store too warm
cashier not helpful	out of ice ^[SEP]
meat tasted bad	telephone out of order
ATM ate card	overcharged ^[SEP]
slippery floor	rolls stale ^[SEP]
music too loud	bread past sale date

July 13

wrong price on spaghetti	undercharged ^[SEP]
water on floor ^[SEP]	out of brown rice ^[SEP]
store looked messy	out of mushrooms
store too warm	overcharged ^[SEP]
checkout lines too long	checkout wait too long
cashier not friendly	shopping cart broken
out of Cheese Doodles	couldn't find aspirin ^[SEP]
triple charged ^[SEP]	out of Tip Top lunch bags
out of Saran Wrap ^[SEP]	out of Tip Top straws
out of Dove Bars	

July 20

out of cucumbers	out of Tip Top toilet paper
checkout lines too slow	out of red peppers ^{{L}{SEP}}
found keys in parking lot	out of Tip Top napkins
lost keys ^{{L}{SEP}}	out of apricots
wrong price on sale item	telephone out of order
overcharged on corn	out of cocktail sauce
wrong price on baby food	water on floor ^{{L}{SEP}}
out of 18 oz. Tide ^{{L}{SEP}}	out of onions
out of Tip Top tissues	out of squash ^{{L}{SEP}}
checkout lines too long	out of iceberg lettuce ^{{L}{SEP}}
out of romaine lettuce	out of Tip Top paper towels

July 27

out of bananas ^{{L}{SEP}}	wanted to know who won the lottery
reported accident in parking lot	store too warm ^{{L}{SEP}}
wrong price on cranapple juice	oatmeal spilled in bulk section
out of carrots ^{{L}{SEP}}	telephone out of order
out of fresh figs ^{{L}{SEP}}	out of Tip Top tissues
out of Tip Top napkins ^{{L}{SEP}}	water on floor ^{{L}{SEP}}
out of Tip Top straws ^{{L}{SEP}}	out of Tip Top paper towels
windows dirty ^{{L}{SEP}}	out of Tip Top toilet paper
out of iceberg lettuce ^{{L}{SEP}}	spaghetti sauce on floor
dislike store decorations ^{{L}{SEP}}	out of Peter Pan crunchy peanut butter
out of Tip Top lunch bags ^{{L}{SEP}}	
out of vanilla soy milk	

Quality Control

- **TOYS, INC.**

Toys, Inc., is a 20-year-old company engaged in the manufacture and sale of toys and board games. The company has built a reputation on quality and innovation. Although the company is one of the leaders in its field, sales have leveled off in recent years. For the most recent six-month period, sales actually declined compared with the same period last year. The production manager, Ed Murphy, attributed the lack of sales growth to “the economy.” He was prompted to undertake a number of belt-tightening moves that included cuts in production costs and layoffs in the design and product development departments. Although profits are still flat, he believes that within the next six months, the results of his decisions will be reflected in increased profits.

The vice president of sales, Joe Martin, has been concerned with customer complaints about the company’s realistic line of working-model factories, farms, and service stations. The moving parts on certain models have become disengaged and fail to operate or operate erratically. His assistant, Keith McNally, has proposed a trade-in program by which customers could replace malfunctioning models with new ones. McNally believes that this will demonstrate goodwill and appease dissatisfied customers. He also proposes rebuilding the trade-ins and selling them at discounted prices in the company’s retail outlet store. He doesn’t think that this will take away from sales of new models. Under McNally’s program, no new staff would be needed. Regular workers would perform needed repairs during periods of seasonal slowdowns, thus keeping production level.

When Steve Bukowski, a production assistant, heard Keith’s proposal, he said that a better option would be to increase inspection of finished models before they were shipped. “With 100 percent inspection, we can weed out any defective models and avoid the problem entirely.”

Take the role of a consultant who has been called in for advice by the company president, Marybeth Corbella. What do you recommend?

- **TIGER TOOLS**

Tiger Tools, a division of Drillmore Industries, was about to launch a new product. Production Manager Michelle York asked her assistant, Jim Peterson, to check the capability of the oven used in the process. Jim obtained 18 random samples of 20 pieces each. The results of those samples are shown in the following table. After he analyzed the data, he concluded that the process was not capable based on a specification width of 1.44 cm.

Sample	Mean	Range	Sample	Mean	Range
1	45.01	.85	10	44.97	.91
2	44.99	.89	11	45.11	.84
3	45.02	.86	12	44.96	.87
4	45.00	.91	13	45.00	.86
5	45.04	.87	14	44.92	.89
6	44.98	.90	15	45.06	.87
7	44.91	.86	16	44.94	.86
8	45.04	.89	17	45.00	.85
9	45.00	.85	18	45.03	.88

Michelle was quite disappointed when she heard this. She had hoped that with the introduction of the new product her operation could run close to full capacity and regain some of its lost luster. The company had a freeze on capital expenditures of more than \$10,000, and a replacement oven would cost many times that amount. Jim Peterson worked with the oven crew to see if perhaps different settings could produce the desired results, but they were unable to achieve any meaningful improvements.

Still not ready to concede, Michelle contacted one of her former professors and explained the problem. The professor suggested obtaining another set of samples, this time using a smaller sample size and taking more samples. Michelle then conferred with Jim and they agreed that he would take 27 samples of five observations each. The results are shown in the following table.

Sample	Mean	Range	Sample	Mean	Range
1	44.96	.42	15	45.00	.39
2	44.98	.39	16	44.95	.41
3	44.96	.41	17	44.94	.43
4	44.97	.37	18	44.94	.40
5	45.02	.39	19	44.87	.38
6	45.03	.40	20	44.95	.41
7	45.04	.39	21	44.93	.39
8	45.02	.42	22	44.96	.41
9	45.08	.38	23	44.99	.40
10	45.12	.40	24	45.00	.44
11	45.07	.41	25	45.03	.42
12	45.02	.38	26	45.04	.38
13	45.01	.41	27	45.03	.40
14	44.98	.40			

• **Questions:** *Consider the following questions, and then write a brief report to Michelle summarizing your findings.*

1. How did Jim conclude that the process was not capable based on his first set of samples? (Hint: Estimate the process standard deviation, σ , using $A_2\bar{R} \approx 3 \frac{\sigma}{\sqrt{n}}$.)
2. Does the second set of samples show anything that the first set did not? Explain what and why.
3. Assuming the problem can be found and corrected, what impact do you think this would have on the capability of the process? Compute the potential process capability using the second data set.
4. If small samples can reveal something that large samples might not, why not just take small samples in every situation?

Inventory Management

• UPD MANUFACTURING

UPD Manufacturing produces a range of health care appliances for hospital as well as for home use. The company has experienced a steady demand for its products, which are highly regarded in the health care field. Recently the company has undertaken a review of its inventory ordering procedures as part of a larger effort to reduce costs.

One of the company's products is a blood pressure testing kit. UPD manufactures all of the components for the kit in-house except for the digital display unit. The display units are ordered at six-week intervals from the supplier. This ordering system began about five years ago, because the supplier insisted on it. However, that supplier was bought out by another supplier about a year ago, and the six-week ordering requirement is no longer in place. Nonetheless, UPD has continued to use the six-week ordering policy. According to purchasing manager Tom Chambers, "Unless somebody can give me a reason for changing, I'm going to stick with what we've been doing. I don't have time to reinvent the wheel."

Further discussions with Tom revealed a cost of \$32 to order and receive a shipment of display units from the supplier. The company assembles 89 kits a week. Also, information from Sara James, in Accounting, indicated a weekly carrying cost of \$.08 for each display unit.

The supplier has been quite reliable with deliveries; orders are received five working days after they are faxed to the supplier. Tom indicated that as far as he was concerned, lead-time variability is virtually nonexistent.

• *Questions*

1. Would using an order interval other than every six weeks reduce costs? If so, what order interval would be best, and what order size would that involve?
2. Would you recommend changing to the optimal order interval? Explain.

• HARVEY INDUSTRIES

Background

Harvey Industries, a Wisconsin company, specializes in the assembly of highpressure washer systems and in the sale of repair parts for these systems. The products range from small portable highpressure washers to large industrial installations for snow removal from vehicles stored outdoors during the winter months. Typical uses for high-pressure water cleaning include:

Automobiles ^{[L][SEP]}	Airplanes ^{[L][SEP]}
Building maintenance	Barns ^{[L][SEP]}
Engines ^{[L][SEP]}	Ice cream plants
Lift trucks ^{[L][SEP]}	Machinery
Swimming pools	

Industrial customers include General Motors, Ford, Chrysler, Delta Airlines, United Parcel Service, and Shell Oil Company.

Although the industrial applications are a significant part of its sales, Harvey Industries is primarily an assembler of equipment for coin operated self-service car wash systems. The typical car wash is of concrete block construction with an equipment room in the center, flanked on either side by several bays. The cars are driven into the bays where the owner can wash and wax the car, utilizing high-pressure hot water and liquid wax. A dollar bill changer is available to provide change for the use of the equipment and the purchase of various products from dispensers. The products include towels, tire cleaner, and upholstery cleaner.

Current Inventory Control System

The current inventory control “system” consists of orders for stock replenishment being made by the stockroom foreman, the purchasing manager, or the manufacturing manager whenever one of them notices that the inventory is low. An order for replenishment of inventory is also placed whenever someone (either a customer or an employee in the assembly area) wants an item and it is not in stock.

Some inventory is needed for the assembly of the high-pressure equipment for the car wash and industrial applications. There are current and accurate bills of material for these assemblies. The material needs to support the assembly schedule are generally known well in advance of the build schedule.

Many inventory transactions are for repair parts and for supplies used by the car washes, such as paper towels, detergent, and wax concentrate. Because of the constant and rugged use of the car wash equipment, there is a steady demand for the various repair parts.

The stockroom is well organized, with parts stored in locations according to each vendor. The number of vendors is relatively limited, with each vendor generally supplying many different parts. For example, the repair parts from Allen Bradley, a manufacturer of electrical motors, are stocked in the same location. These repair parts will be used to provide service for the many electrical motors that are part of the high-pressure pump and motor assembly used by all the car washes.

In recent years Harvey Industries has been in financial difficulty. The company has lost money for three of the last four years, with the last year’s loss being \$17,174 on sales of \$1,238,674. Inventory levels have been steadily increasing to their present levels of \$124,324.

The company employs 23 people with the management team consisting of the following key employees: president, sales manager, manufacturing manager, controller, and purchasing manager. The abbreviated organization chart reflects the reporting relationship of the key employees and the three individuals who report directly to the manufacturing manager.

Because of the heavy sales volume of repair parts, there are generally two employees working in the stockroom—a stockroom foreman who reports to the manufacturing manager and an assistant to the foreman. One of these two employees will handle customer orders. Many customers stop by and order the parts and supplies they need. Telephone orders are also received and are shipped by United Parcel Service the same day.

The assembly area has some inventory stored on the shop floor. This inventory consists of low-value items that are used every day, such as nuts, bolts, screws, and washers. These

purchased items do not amount to very much dollar volume throughout the year. Unfortunately, oftentimes the assembly area is out of one of these basic items and this causes a significant amount of downtime for the assembly lines.

Paperwork is kept to a minimum. A sales slip listing the part numbers and quantities sold to a customer is generally made out for each sale. If the assembly department needs items that are not stocked on the assembly floor, someone from that department will enter the stockroom and withdraw the necessary material. There is no paperwork made out for the items needed on the assembly floor.

There were 973 different part numbers purchased for stock last year and those purchases amounted to \$314,673. An analysis of inventory records shows that \$220,684 was spent on just 179 of the part numbers.

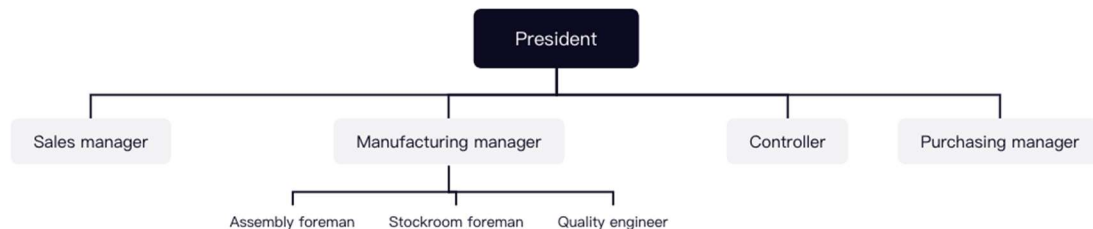
Fortunately for Harvey Industries, most of the items they purchase are stocked by either the manufacturer or by a wholesaler. When it is discovered that the company is out of stock on an item, it generally takes only two or three days to replenish the stock.

Due to the company's recent losses, its auditing firm became concerned about the company's ability to continue in business. Recently the company sold off excess vacant land adjoining its manufacturing facility to generate cash to meet its financial obligations.

New President

Because of the recent death of the owner, the trust department of a Milwaukee bank (as trustee for the state) has taken over the company's affairs and has appointed a new company president. The new president has identified many problem areas—one of which is improper inventory control. He has retained you as a consultant to make specific recommendations concerning a revised inventory control system. What are your recommendations and their rationale?

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• GRILL RITE

Grill Rite is an old-line company that started out making wooden matches. As that business waned, the company entered the electric barbecue grill market, with five models of grills it sells nationally. For many years the company maintained a single warehouse from which it supplied its distributors.

The plant where the company produces barbecue sets is located in a small town, and many workers have been with the company for many years. During the transition from

wooden matches to barbecue grills, many employees gave up their weekends to help with changing over the plant and learning the new skills they would need, without pay. In fact, Mac Wilson, the company president, can reel off a string of such instances of worker loyalty. He has vowed to never lay off any workers, and to maintain a full employment, steady rate of output. “Yes, I know demand for these babies (barbecue grills) is seasonal, but the inventory boys will just have to deal with it. On an annual basis, our output matches sales.”

Inventory is handled by a system of four warehouses. There is a central warehouse located near the plant that supplies some customers directly, and the three regional warehouses.

The vice president for sales, Julie Berry, is becoming increasingly frustrated with the inventory system that she says “is antiquated and unresponsive.” She points to increasing complaints from regional sales managers about poor customer service, saying customer orders go unfilled or are late, apparently due to shortages at the regional warehouse. Regional warehouse managers, stung by complaints from sales managers, have responded by increasing their order sizes from the main warehouse, and maintaining larger amounts of safety stock. This has resulted in increased inventory holding costs, but it hasn’t eliminated the problem. Complaints are still coming in from salespeople about shortages and lost sales. According to managers of the regional warehouses, their orders to the main warehouse aren’t being shipped, or when they are, they are smaller quantities than requested. The manager of the main warehouse, Jimmy Joe (“JJ”) Sorely, says his policy is to give preference to “filling direct orders from actual customers, rather than warehouse orders that might simply reflect warehouses trying to replenish their safety stock. And besides, I never know when I’ll get hit with an order from one of the regional warehouses. I guess they think we’ve got an unlimited supply.” Then he adds, “I thought when we added the warehouses, we could just divide our inventory among the warehouses, and everything would be okay.”

When informed of the “actual customers” remark, a regional warehouse manager exclaimed, “We’re their biggest customer!”

Julie Berry also mentioned that on more than one occasion she has found that items that were out of stock at one regional warehouse were in ample supply in at least one other regional warehouse.

Take the position of a consultant called in by president Mac Wilson. What recommendations can you make to alleviate the problems the company is encountering?

• FARMERS RESTAURANT

SARAH LUBBERS AND CHRIS RUSCHE, GRAND VALLEY STATE UNIVERSITY^[L]_[SEP]

Farmers Restaurant is a full-service restaurant offering a variety of breakfast, lunch, and dinner items. Currently, Kristin Davis is the general manager for the Farmers Restaurant located in the Grand Rapids/ Wyoming metro area of Michigan. Since becoming manager, Kristin has faced some difficulties with ordering the right amounts of food items for the restaurant. Because of this, there are some weeks the restaurant has a surplus of menu items that are no longer fresh, and must be discarded. At other times, the restaurant has experienced shortages of some items.

The fact that inventory accounts for an average cost of 26 percent of the restaurant's total revenues underscores the importance of managing inventory. Kristin would like to find a way to ensure that she is maintaining the proper amount of inventory. Customer counts at Kristin's restaurant have been declining recently, so one of Kristin's greatest focuses is to keep current customers and attract new customers. She believes that a key aspect of this is having all of the items on the menu in stock.

The restaurant industry is competitive. In the Grand Rapids/ Wyoming metro area alone there are over 1,600 restaurants. Some of Farmers Restaurant's most serious competitors are IHOP, Applebee's, and Big Boy, all of which are located within 20 miles of the Farmers Restaurant, so customers have many alternatives from which to choose.

Online inventory systems are used to assist restaurant managers in determining on-hand inventory and gauging how well the restaurant is controlling food costs. The fiscal week for Farmers Restaurant starts on Thursday and ends on Wednesday of the following week. Each Wednesday, the manager physically counts the inventory on hand and enters the data into the online inventory system. The computer software system then compares the on-hand inventory for that week, the amount of food ordered, and the inventory on hand for the end of the previous week with the sales for the current week. By doing so, it is able to determine a total food cost. The manager compares this cost with the benchmark cost to see how well the restaurant has been managing its inventory. This is one of the most important numbers to managers at the Farmers Restaurant because it accounts for approximately 30 percent of total costs in terms of a store's cost structure.

The computer software system also compares the total cost of food on hand with the total amount of sales for that week and computes a percentage of on-hand inventories. As a guideline, the company has set a standard of having between 29 and 36 percent for its on-hand inventory level. The company feels that this level of inventory is an appropriate average to ensure quality food that is fresh and within expiration. Lastly, it is better to keep the inventory at a minimum level to ensure the accuracy and ease of inventory counts.

The Farmers Restaurant Kristin manages has been running above average in terms of food costs. For this reason, her boss has become concerned with the performance of the ordering system she is using at her restaurant. Kristin has been using her intuition to decide how much product to order despite the fact that the product order sheets provide a moving average usage of each product. Kristin bases her inventory management on her intuition because she does not understand how to utilize the moving average forecasting technique when placing orders. An additional complication with ordering inventory is that each item is packed in multiple quantities, so she cannot order the exact amount that she needs. Her boss requested that she create a more accurate way of ordering food and to report back to him in one month. Kristin is worried that if she cuts inventory levels too low she will run out of products, which may result in a decrease in customer counts.

After Kristin met with her boss, she began to think about what changes she could

make. She knows that inventory has been a weak point for her, but she remembers one of her employees talking about inventory management from one of his college courses. Kristin decides to ask the employee if he would be willing to help her try and come up with a better way for her to order products. Kristin tells him how the ordering system works, shows him the ordering form, and relates the given information.

Suppose you have been asked to work with Kristin to improve inventory ordering.

• **Questions**

1. Describe the importance of inventory management as it relates to the Farmers Restaurant.
2. What ordering system would be best for this situation?^[L]_[SEP]
3. Given the following information, provide an example of how much of Farmers Sausage Gravy Mix should be ordered. You are doing the order for Thursday. Also, Kristin would like a service level of 95 percent, and you have found that there is a standard deviation of 3.5 units per week, and a moving average weekly demand of 35 servings. The gravy mix comes in packs of two servings. There are currently three packs in inventory.
4. Given the above information and an on-hand inventory of 12, determine the risk of stock out at the end of initial lead time and at the end of the second lead time. The lead time is 2 days and orders are placed once a week.
5. The supplier Kristin uses is in Ohio. Why might Kristin consider dealing with a nearby supplier instead of the one^[L]_[SEP] in Ohio? What reasons might there be for not switching suppliers?